# Newton's LAWS OF MOTION

## FORCE :-

Force is a push or pull which changes or tries to change the state of rest, the state of uniform motion, size or shape of a body, direction of motion.

## Key points :-

- Its SI units is newton (N) and its dimensional formula is $[MLT^{-2}]$

## Units for measurement of force :-

| Absolute units | Other units |
|---|---|
| i) N (MKS) | kg-wt or kg-f (kg-force) |
| ii) dyne (C.G.S.) | g-wt or g-f (g-force) |

## Relation between above units :-

i) 1 kg-wt = 9.8 N = 1 kg-f

ii) 1 g-wt = 980 dyne = 1 g-f

iii) 1 N = $10^5$ dyne

INERTIA: - The property of an object by virtue of which it can not change its state of rest or of uniform motion along a straight line on its own, is called inertia.

Key points :-

- Inertia is a measure of mass of a body. Greater the mass of a body greater will be its inertia or vice-versa.
- Inertia has no units and no dimensions.
- Two bodies of equal mass, one in motion and another is at rest, possess same inertia because it is a factor of mass only and does not depend upon the velocity.

## Types of Inertia :-

(i) Inertia of rest :- It is defined as the property of a body to resist any change in its state of rest.

When a bus or train starts to move suddenly, the passengers sitting inside it falls backward due to inertia of rest.

(ii) Inertia of motion :- It is defined as the property of a body to resist any change to its state of uniform motion.

When a moving bus or train stops suddenly, the passengers sitting in it jerks in forward direction due to inertia of motion.

(iii) **Inertia of Direction**:- It is defined as a body's ability to resist changes in its motion direction.

When a bus turns a corner, passengers tend to lean or slide towards the opposite side, as their bodies want to continue moving on a straight line.

## LINEAR MOMENTUM:-

- The total quantity of motion possessed by a moving body is known as the momentum of the body.
- It is the product of the mass and velocity of a body.

Mathematically, $\boxed{\vec{p} = m\vec{v}}$

- SI unit - kg-m/s  Dimension - $[MLT^{-1}]$

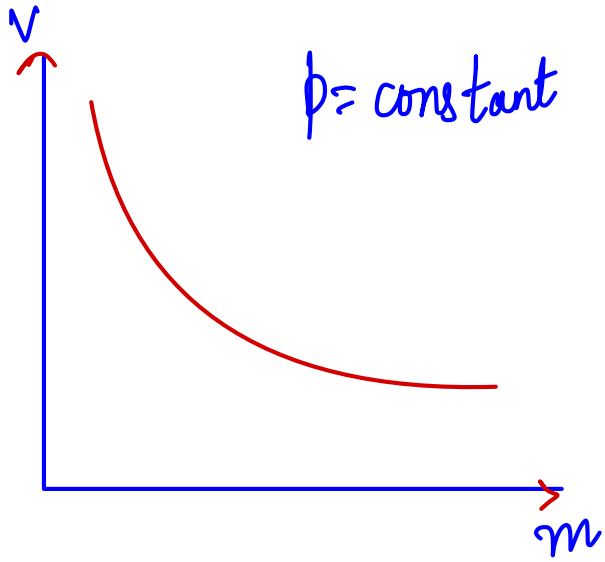
  CGS unit - g-cm/s

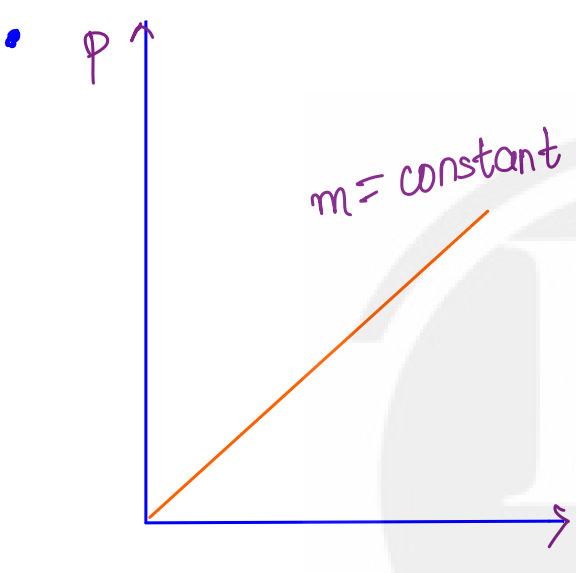
- It is a vector quantity and its direction is in the direction of velocity of the body.
- If two objects of different masses have same momentum, the lighter body possesses greater velocity

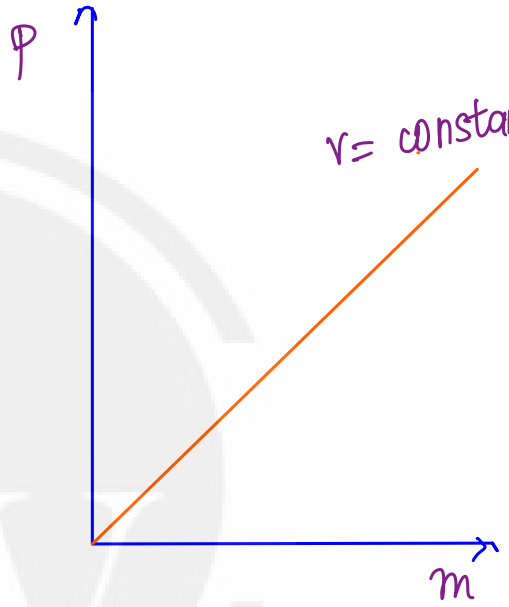
$p = m_1v_1 = m_2v_2 = \text{constant}$

$$\frac{V_1}{V_2} = \frac{m_2}{m_1}$$

i.e. $V \propto \frac{1}{m}$




# Newton's Laws Of Motion :

## Newton's First Law (Law of Inertia)

A body continues to be in state of rest or in uniform motion along a straight line unless an external force is applied on it.

- If no net force acts on a body, then the velocity of the body can not change i.e., the body can not accelerate

- This law defines inertia and that's why also known as laws of inertia.

# Newton's Second Law:

The rate of change of linear momentum is proportional to the applied force in the direction of applied force.

# Newton's second law of motion is called real law of motion because first and third law of motion can be obtained by it.

Mathematically,

$$\vec{F} = \frac{\Delta \vec{P}}{\Delta t} \quad \text{or} \quad \vec{F} = \frac{d\vec{P}}{dt}$$

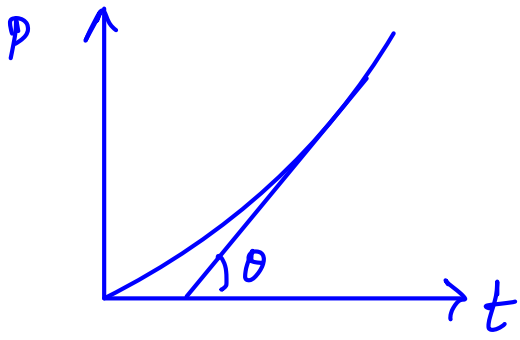
Or $$\vec{F} = \frac{d}{dt}(m\vec{v}) = m\frac{d\vec{v}}{dt} = m\vec{a}$$

$$\therefore \boxed{\vec{F} = m\vec{a}}$$

$\{ 1\ N = 1\ kg\text{-}m/s^2 \}$

$1\ dyne = 1\ g\text{-}cm/s^2$

**Note:-** The slope of momentum - time graph is equal to the force on the particle at that instant.




$$F = \frac{dp}{dt} = \text{slope} = \tan\theta$$

Note :- $\vec{F} = m\vec{a}$ formula is valid only if the mass of the body is constant and finite

If m is not constant

$$\vec{F} = \frac{d}{dt}(m\vec{v}) = m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt}$$

- If force and acceleration have three components along x, y and z axis, then

$$\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k} \quad \text{and} \quad \vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$

$$F_x = ma_x, \; F_y = ma_y, \; F_z = ma_z$$

- No force is required to move a body uniformly along a straight line with constant speed.

$$\vec{F} = 0 \quad (\text{as } \vec{a} = 0)$$

## Newton's Third Law:

For every action, there is an equal and opposite reaction and both acts on two different bodies

If $\vec{F}_{AB}$ = force exerted on body A by body B (Action) and $\vec{F}_{BA}$ = force exerted on body B by body A (Reaction)

Then, $\boxed{\vec{F}_{AB} = -\vec{F}_{BA}}$

Examples -

- Swimming is possible due to third law of motion.
- When a gun is fired, the bullet moves forward (action). The gun recoils backward (reaction)
- While walking, a person presses the ground in the backward direction by his feet (action) and ground pushes the person in forward direction with an equal force (reaction)

## EQUILIBRIUM:

It can be defined as a state where all forces acting on a body are balanced out, resulting in a net force of zero. ($\vec{a} = 0$)

When a body is in equilibrium, it does not experience a change in its **state of rest or uniform motion**

## FREE BODY DIAGRAM:

In FBD, the object of interest is isolated from its surroundings and the interactions between the object and the surroundings are represented in terms of forces.

# Common forces:-

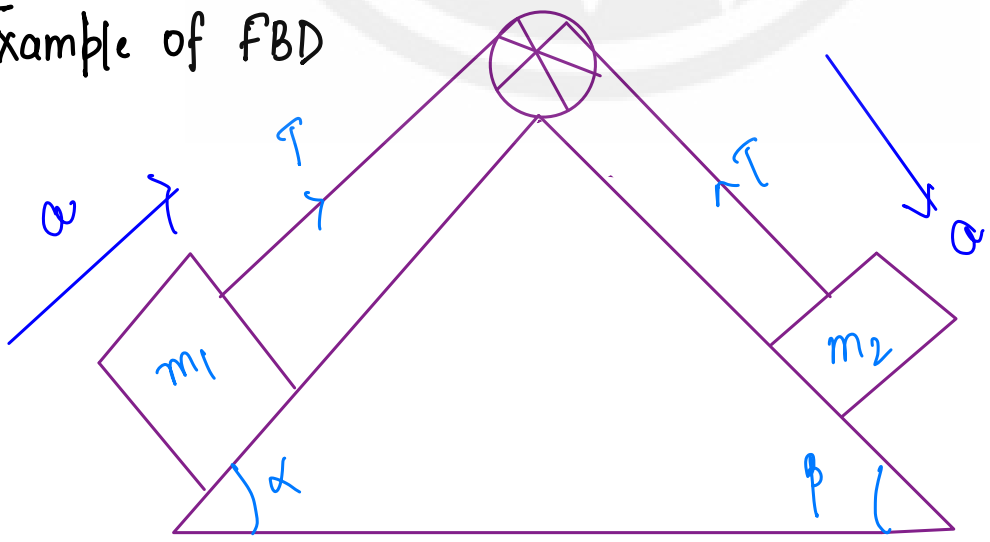
**Weight:-** • Acts downward
• $W = mg$, where $m$ = mass, $g = 9.8\ m/s^2$

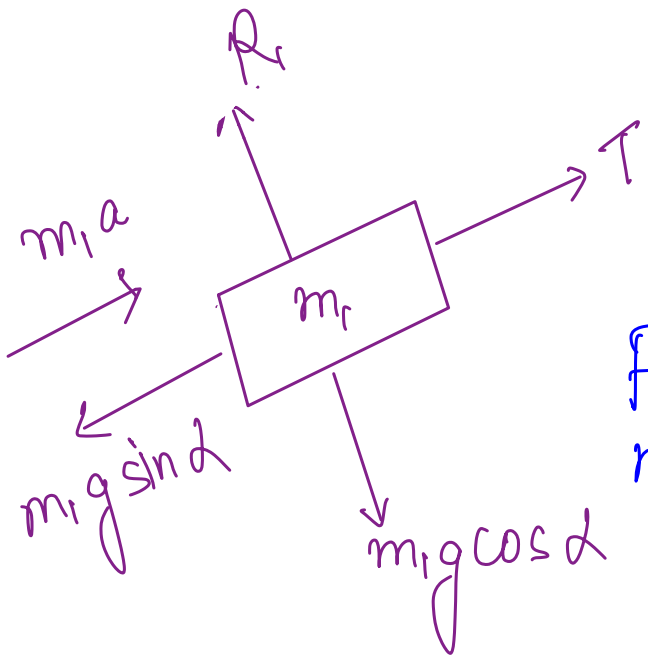
**Normal force:-** (or Reaction) (R)
• When a body is placed on a rigid surface, the body experiences a force which is perpendicular to the surfaces in contact.

**Tension (T):-**
• Force through a string, rope or cable.
• Acts away from the object along the string.

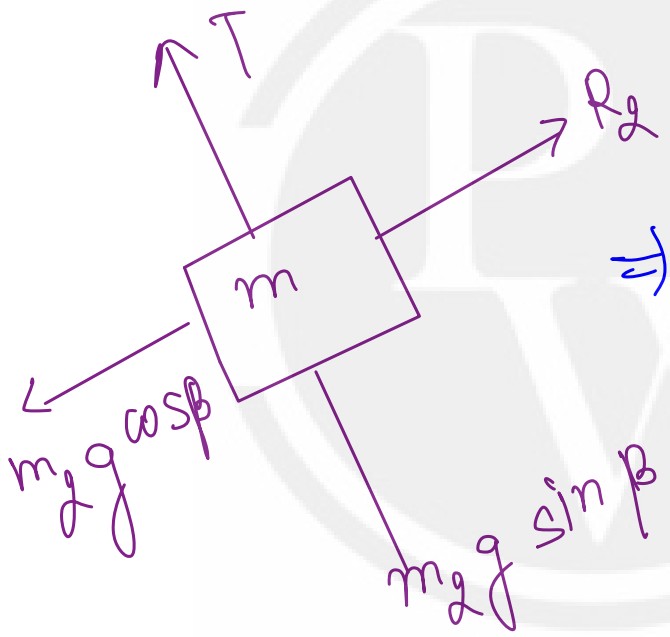
**Friction:** It is the force resisting the relative motion between two surfaces in contact. (will be discussed later in detail)

Example of FBD






$\Rightarrow$ Free body diagram of mass $m_1$




$\Rightarrow$ Free body diagram of mass $m_2$

## Impulsive force :-

When a large force works on a body for a very small time interval, it is called impulsive force.

→ The product of impulsive force and time for which it acts is called impulse.

* Impulse = Force × Time

$$\vec{I} = \int_{t_1}^{t_2} \vec{F}\, dt$$

* Its SI unit is N-S or kg-m/s and its dimension is $[MLT^{-1}]$

* It is a vector quantity and its direction is in the direction of force

* force-time graph :- Impulse is equal to the area under F-t graph.

from newton's 2nd law $\vec{F} = \frac{d\vec{p}}{dt}$

or $\int_{t_1}^{t_2} \vec{F}\, dt = \int_{p_1}^{p_2} d\vec{p}$

or $\boxed{\vec{I} = \vec{p}_2 - \vec{p}_1 = \Delta\vec{p} = \text{change in momentum}}$

⇓

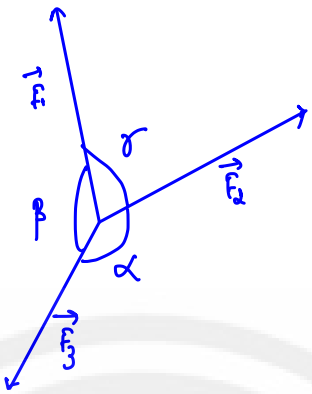
Impulse momentum Theorem

Example - While catching a ball, a player draws his hands **downwards** which increases the time of contact and so lesser force acts on his hands and his hands are saved from getting hurt.

## Lami's Theorem :-

for three concurrent forces in equilibrium,

$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$




## Law of Conservation of Linear Momentum :-

If no external force acts on a system of constant mass, then the total momentum of the system remains constant with time.

We know $\vec{F} = \frac{d\vec{p}}{dt}$

If $\vec{F} = 0$ then $\vec{p}$ = constant

i.e. $\vec{p} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \cdots$ = constant

i.e. $m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3 + \cdots$ = constant

- Law of conservation of linear momentum is independent of frame of reference, though linear momentum depends on frame of reference.

**Recoiling of gun :-** for bullet and gun system, there is no external force, so the momentum of the system remains unaffected.

Initial momentum of system = 0

final momentum of system = $m_G \vec{v}_G + m_B \vec{v}_B$

$\therefore \quad m_G \vec{v}_G + m_B \vec{v}_B = 0$ { $m_G$ : mass of gun, $m_B$ : mass of bullet }

$\therefore \quad \vec{v}_G = -\frac{m_B}{m_G} \vec{v}_B$ { $v_G$ : velocity of gun, $v_B$ : velocity of bullet }

* -ve sign indicates that gun moves in opposite direction to the bullet.

Explosions :- Let a body of mass M at rest explods into two parts of masses $m_1$ & $m_2$

Before explosion, $\vec{p}_i = 0$

After explosion, $\vec{p}_f = m_1 \vec{v}_1 + m_2 \vec{v}_2$

$\therefore \vec{F}_{ext} = 0 \Rightarrow m_1 \vec{v}_1 + m_2 \vec{v}_2 = 0 \Rightarrow \vec{v}_2 = -\frac{m_1}{m_2} \vec{v}_1$

# Frame of Reference :-

A frame of reference is a coordinate system or a set of axes within which the position, velocity and acceleration of object is measured

Types of frame of reference :-

(i) Inertial frame of reference :- A frame that is at rest or moves with constant velocity.

- Newton's laws hold true in this frame.

(ii) Non-Inertial frame of reference :- Accelerated frame of references are called non-inertial frame of reference.

- Newton's laws do not apply directly unless pseudo-forces are introduced.

Example - a rotating merry go-round or an accelerating car.

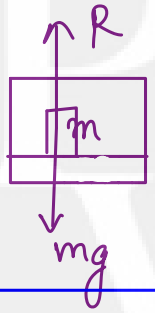
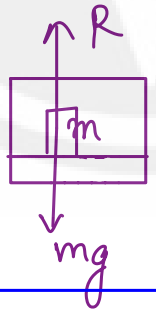
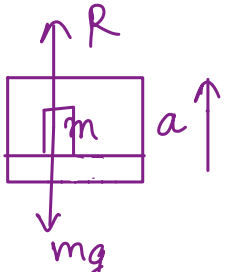
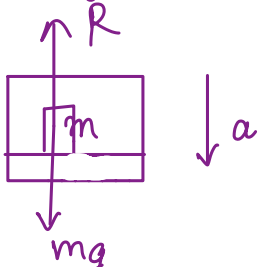
**Pseudo force :-** It is an apparent force that arises when an observer is in a non-inertial frame of reference.

→ It does not result from any physical interaction but is introduced to apply Newton's laws in an accelerating frame.

$$\vec{F}_{pseudo} = -m\vec{a}$$

- $m$ = mass of the object
- $\vec{a}$ = acceleration of the non-inertial frame.

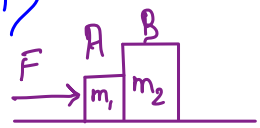
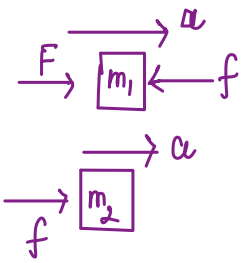
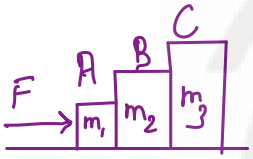
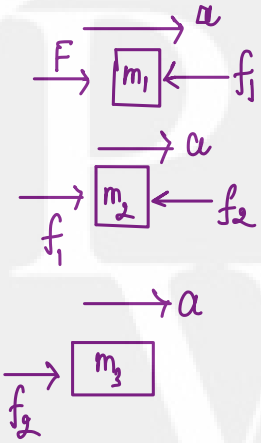
## Apparent Weight of a Body in a Lift:-

| Condition | Diagram | Apparent weight (R) |
|---|---|---|
| 1) Lift is at rest |   $v=0$, $a=0$ | $R-mg=0$ <br> $\Rightarrow R = mg$ |
| 2) Lift moving upward or downward with constant velocity |   $v=$ constant, $a=0$ | $R-mg=0$ <br> $R = mg$ |
| 3) Lift accelerating upward with acceleration 'a' |   | $R-mg = ma$ <br> $\therefore R = m(g+a)$ |
| 4) Lift accelerating downward with with the acceleration 'a' |   | $mg-R = ma$ <br> $R = m(g-a)$ |

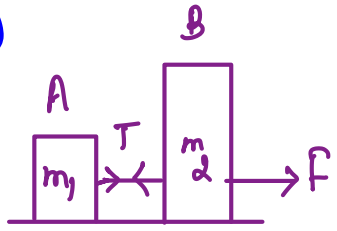
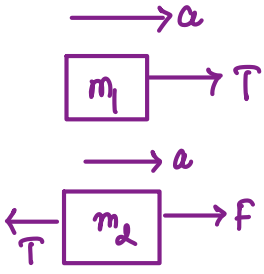
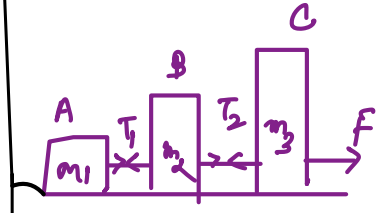
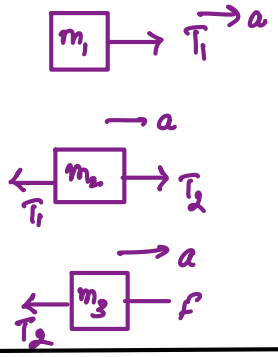
* In a lift in free fall, a person apparent weight becomes zero as ($a=g$), meaning they experience weightlessness.

# Calculation of Acceleration :-

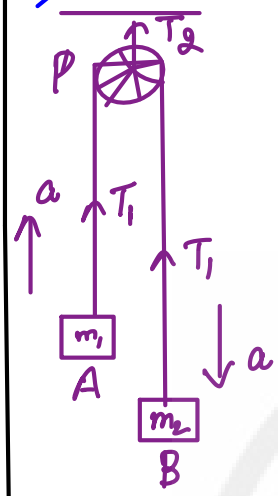
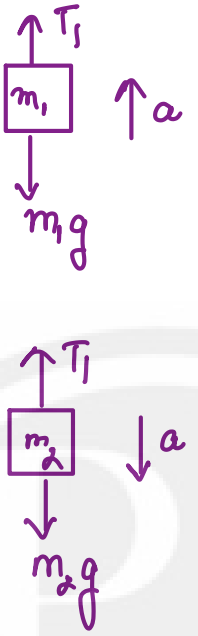
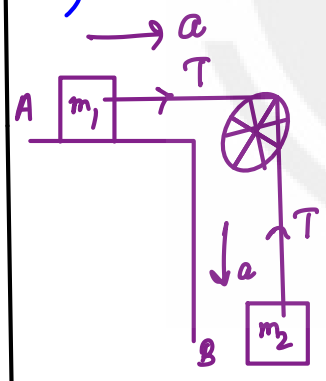
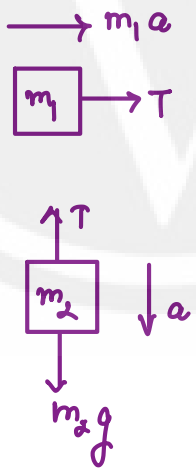
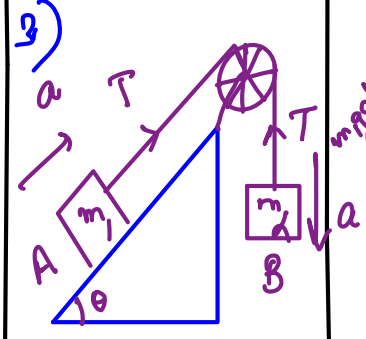
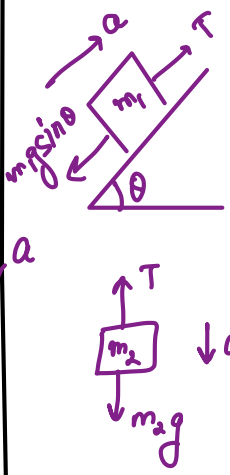
## Motion of Blocks in contact :-

| Condition | FBD | Equation | Force and acceleration |
|---|---|---|---|
| 1.)   (F: applied force) |   | $F-f = m_1 a$ <br> $f = m_2 a$ <br> (f: contact force) | $a = \frac{F}{m_1+m_2}$ <br> $f = \frac{m_2 F}{m_1+m_2}$ |
| 2)   |   | $F-f_1 = m_1 a$ <br> $f_1-f_2 = m_2 a$ <br> $f_2 = m_3 a$ | $a = \frac{F}{m_1+m_2+m_3}$ <br> $f_1 = \frac{(m_2+m_3)F}{m_1+m_2+m_3}$ <br> $f_2 = \frac{m_3 F}{m_1+m_2+m_3}$ |

## Motion of Bodies connected by Mass less string

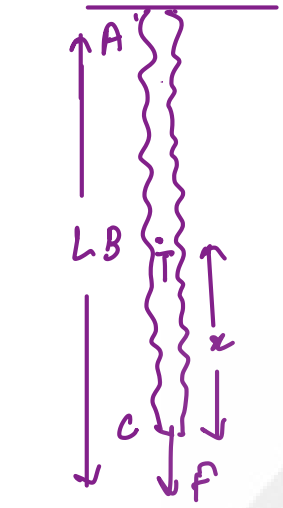
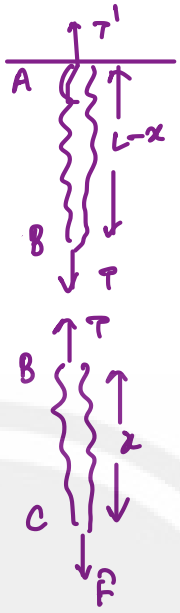
| | | | |
|---|---|---|---|
| 1.)   |   | $T = m_1 a$ <br> $F-T = m_2 a$ | $a = \frac{F}{m_1+m_2}$ <br> $T = \frac{m_1 F}{m_1+m_2}$ |
| 2)   |   | $T_1 = m_1 a$ <br> $T_2-T_1 = m_2 a$ <br> $F-T_2 = m_3 a$ | $a = \frac{F}{m_1+m_2+m_3}$ <br> $T_1 = \frac{m_1 F}{m_1+m_2+m_3}$ <br> $T_2 = \frac{(m_1+m_2)F}{m_1+m_2+m_3}$ |

# Pulley - Block System

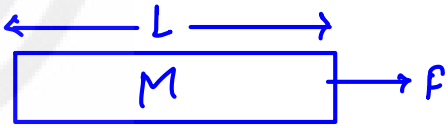
| Condition | FBD | Equation | Tension and acceleration |
|---|---|---|---|
| 1.)   |   | $T_1 - m_1g = m_1a$<br>$m_2g - T_1 = m_2a$<br>$T_2 = 2T_1$ | $T_1 = \left(\frac{2m_1m_2}{m_1+m_2}\right)g$<br>$a = \left[\frac{m_2-m_1}{m_1+m_2}\right]g$ |
| 2.)   |   | $T = m_1a$<br>$m_2a = m_2g - T$ | $a = \left(\frac{m_2}{m_1+m_2}\right)g$<br>$T = \left(\frac{m_1m_2}{m_1+m_2}\right)g$ |
| 3.)   |   | $m_1a = T - m_1g\sin\theta$<br>$m_2a = m_2g - T$<br>$m_2a = m_2g - T$ | $a = \left[\frac{m_2-m_1\sin\theta}{m_1+m_2}\right]g$<br>$T = \frac{m_1m_2(1+\sin\theta)}{m_1+m_2}g$ |

* all surfaces are smooth

# Motion of massive string

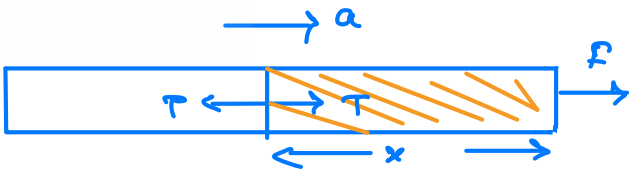
| Condition | FBD | Equations |
|---|---|---|
|   Mass of segment $BC = \left(\frac{M}{L}x\right)$ |   | $T' = \frac{M}{L}(L-x)g + T$<br>$T = F + \frac{M}{L}xg$ |

## Tension in a Rod :-

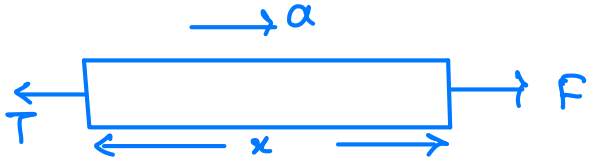



$F - T = ma$

$\Rightarrow T = F - ma$




$\Rightarrow T = F - \frac{mF}{M} \quad \left(\because a = \frac{F}{M}\right)$




Mass of Length $x = \frac{M}{L}x$

$\therefore \quad T = F - \left(\frac{M}{L}x\right)\frac{F}{M} = F\left(1 - \frac{x}{L}\right)$

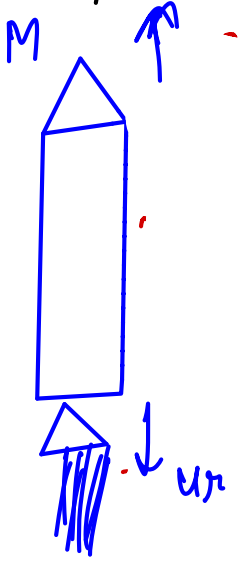
## Rocket Propulsion :-

$M_0$ = initial mass of rocket

$\frac{dm}{dt}$ = rate at which gas is burning

$M_t$ = mass after time t

$u_r$ = velocity of ejecting gases relative to rocket




Thrust on rocket

$$|\vec{F}| = u_r \frac{dM}{dt}$$

Net force on rocket $= u_r \frac{dM}{dt} - Mg$

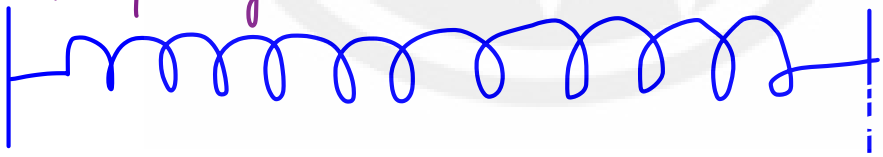
Net acceleration of rocket $= \frac{u_r}{M}\frac{dM}{dt} - g$

where $M = M_0 - \frac{dM}{dt}t$

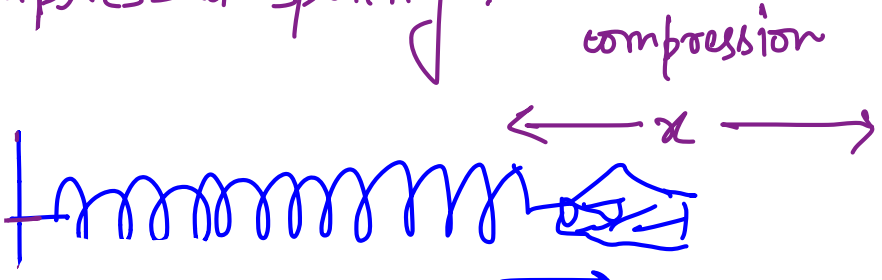
## Spring Force :-

It is the restoring force exerted by a spring when it is stretched or compressed.

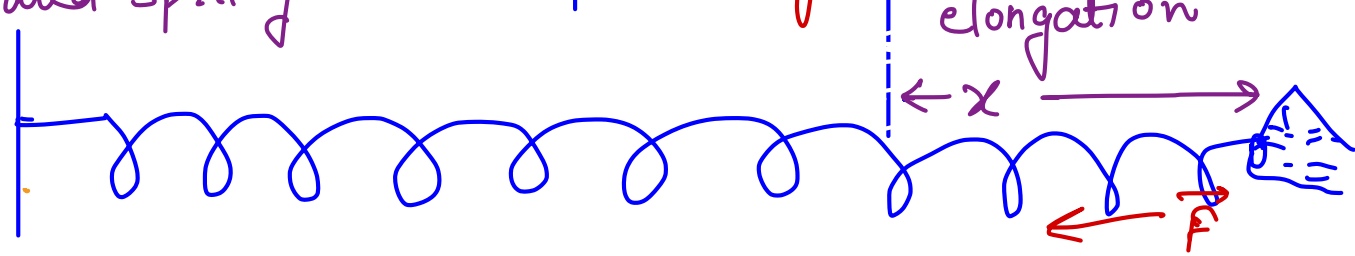
Relaxed spring :-




Compressed spring :-



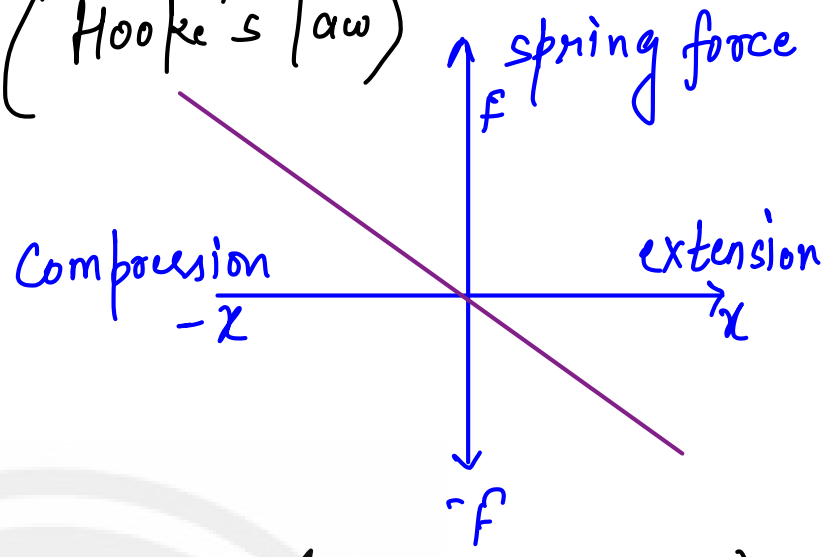

Elongated spring



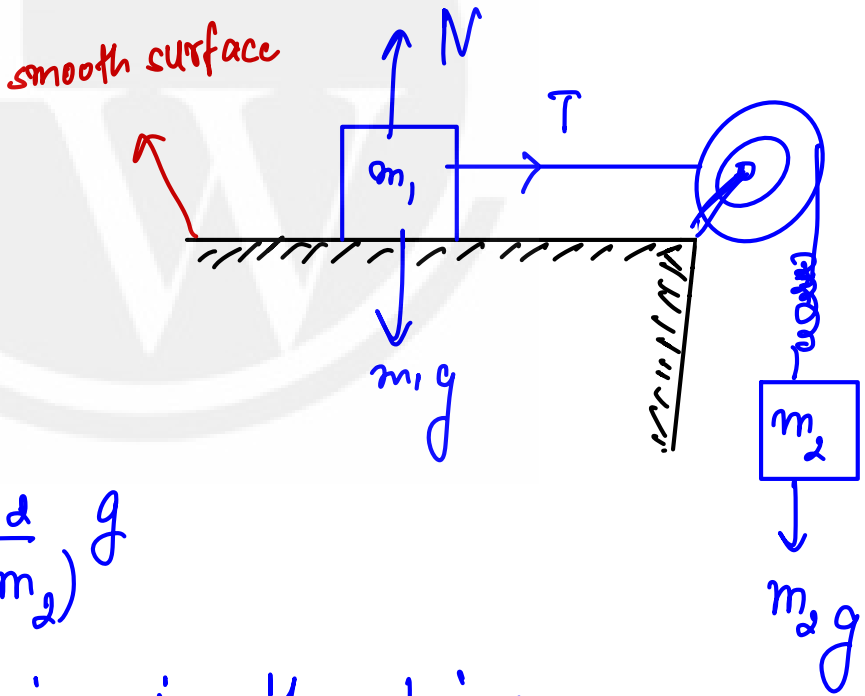
Spring force varies linearly with $x$ and acts in a direction opposite to $x$.

$$F = -kx \quad \text{(Hooke's law)}$$

$k \rightarrow$ spring constant




for a pulley-spring system (at steady state)




acceleration,

$$a = \frac{m_2 g}{m_1 + m_2}$$

Tension, $T = \frac{m_1 m_2}{(m_1 + m_2)} g$

If $x$ is the extension in the spring

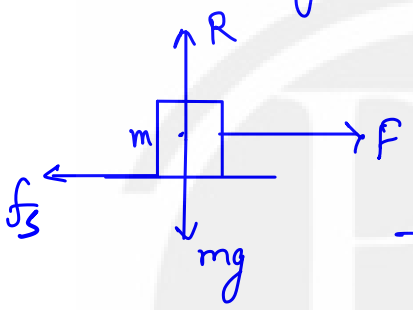
$$T = kx \Rightarrow x = \frac{T}{k} = \frac{m_1 m_2 g}{k(m_1 + m_2)}$$

# FRICTION

force which opposes relative motion / tendency of relative motion.

## Types of friction :-

(i) Static friction :- The opposing force that comes into play when one body tends to move over the surface of another, but the actual motion has not yet started is called static friction.




F → applied force, $f_s$ = force of static friction.

→ If F is applied and body remains at rest $F = f_s$

→ If no pulling force is acting, force of friction is also zero

→ Static friction is self adjusting force.

(ii) limiting friction → If F increases, $f_s$ also increases. If F exceeds a certain (maximum) value, the body starts moving. The maximum value of static friction upto which body does not move is called limiting friction.

**Law of limiting friction :**

$$F_L = \mu_s R \quad \{ \mu_s = \text{coefficient of static friction} \}$$

* frictional force is independent of area in contact.

Note- value of $\mu$ depend on material and nature of surfaces in contact.

(iii) Kinetic friction :- If the applied force is increased further and sets the body in motion, the friction opposing the motion is called kinetic friction.

## Law of kinetic friction :-

$F_k = \mu_k R$ { $\mu_k$ = coefficient of kinetic friction }

Types of kinetic friction :-

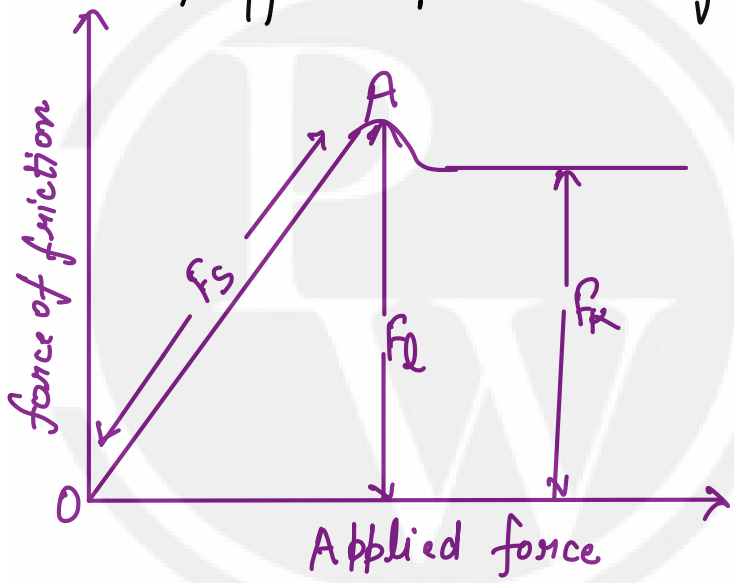
Sliding friction ↓ — when body is actually sliding over the surface

Rolling friction ↓ — when objects rolls over a surface.

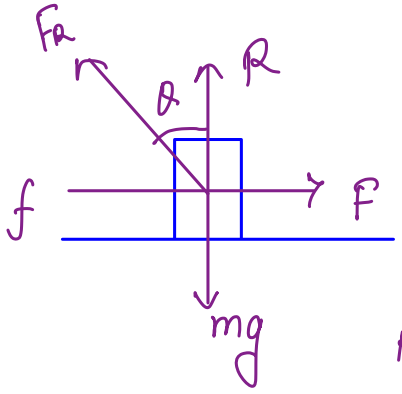
Note :- value of $\mu_s$ and $\mu_k$ depend on the nature of both the surfaces in contact.

→ $\mu_k$ and $\mu_s$ are dimensionless and unitless.

## Graph Between Applied force and force of friction.




**Angle of friction :-** The angle ($\theta$) which the resultant of limiting friction and normal reaction makes with normal reaction.




$$\tan\theta = \frac{f_l}{R} = \mu_s$$

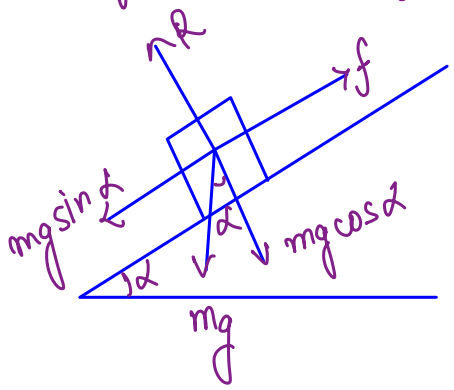
$$\theta = \tan^{-1}(\mu_s)$$

Resultant force exerted by surface on block

$$F_R = \sqrt{f^2 + R^2} = \sqrt{(\mu mg)^2 + (mg)^2}$$

$$= mg\sqrt{\mu^2 + 1}$$

**Angle of Repose ($\alpha$) :-** the angle of inclined plane with horizontal such that a body placed on it just begins to slide



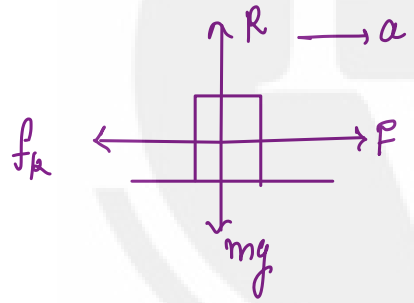

$F = mg\sin\alpha$

$R = mg\cos\alpha$

$\tan\alpha = \frac{f}{R} = \mu_s \Rightarrow \alpha = \tan^{-1}(\mu_s)$

## Acceleration of body on Rough Surface :-

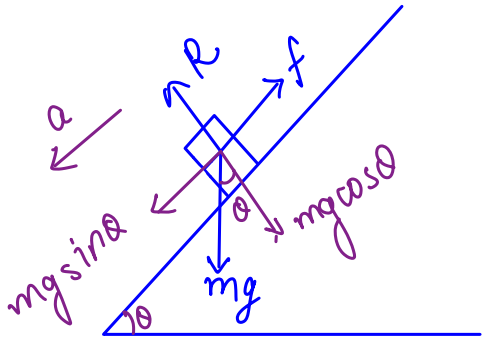
### On horizontal Surface :




$ma = F - f_k$

$a = \frac{F - f_k}{m}$

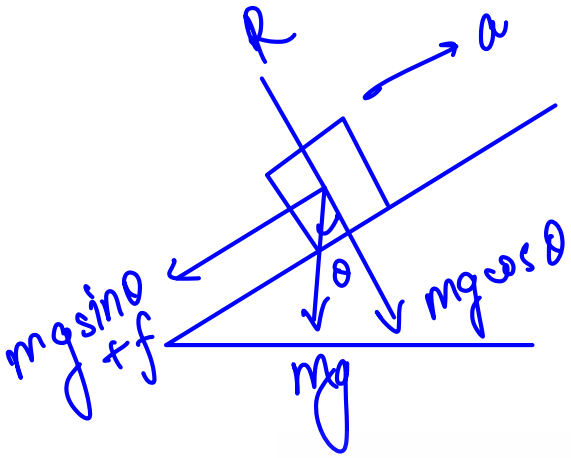
**On Inclined Surface :** When angle plane is more than angle of repose, the body slides down w. teration 'a'.




$ma = mg\sin\theta - f = mg\sin\theta - \mu mg\cos\theta$

$\Rightarrow a = (g\sin\theta - \mu\cos\theta)$

# A Block sliding up a rough inclined plane :-



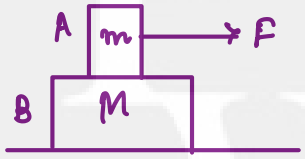

$f = \mu N = \mu mg\cos\theta$

$ma = mg\sin\theta + f$

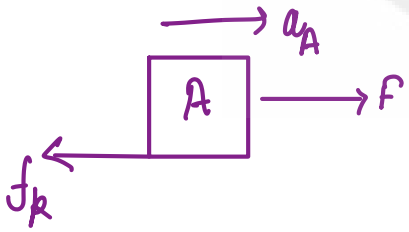
$= mg\sin\theta + \mu mg\cos\theta$

$a = (g\sin\theta + \mu\cos\theta)$

## Motion of two bodies one resting on the other (double block system)




* If friction is present between A & B only $(F > f)$




$F - f_k = m a_A \Rightarrow a_A = \frac{F - f_k}{m}$

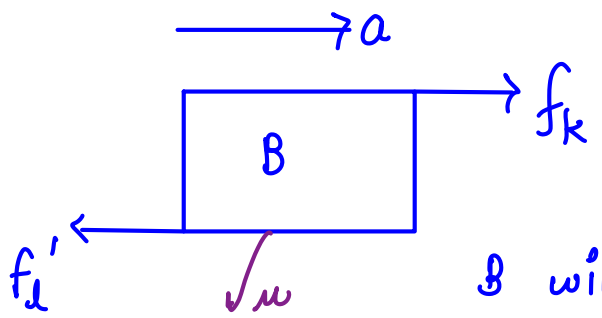
$= \frac{F - \mu_k mg}{m}$




$f_k = M a_B$

$\Rightarrow a_B = \frac{\mu_k mg}{M}$

* If there is a friction between B and floor also




$f_l' = \mu(M+m)g$

B will move only if $f_k > f_l'$

$f_k - f_l' = M a_B$

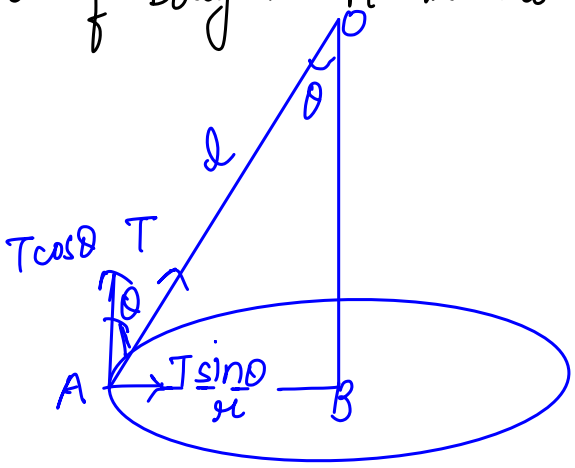
# Dynamics of Circular Motion :

**Centripetal force :-** Centripetal force is the force required to keep a body moving in a circular path with uniform speed. It is directed towards the center of the circular path.

$$F_C = \frac{mv^2}{r}$$

**Centrifugal force :-** It is pseudo force that appears to act on a body when it is observed from a rotating (non-inertial) frame of reference. It acts outward from the center of the circle.

## Motion of body in Horizontal Circle (conical pendulum)




$T\sin\theta = \frac{mv^2}{r}$ ---- (i)

$T\cos\theta = mg$ ---- (ii)

$\tan\theta = \frac{v^2}{rg} \Rightarrow \theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$

$$\left[T = mg\sqrt{1+\frac{v^4}{g^2r^2}}\right]$$

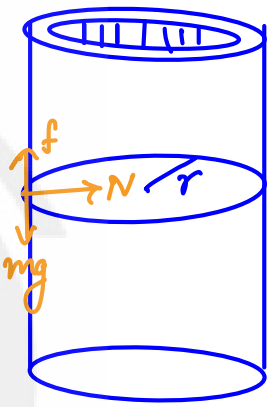
or $\tan\theta = \frac{\omega^2 r}{g}$ ---(1) from diagram $\tan\theta = \frac{r}{l\cos\theta}$ -(2)

from eq (1) & (2) $\omega^2 = \frac{g}{l\cos\theta} \Rightarrow \omega = \sqrt{\frac{g}{l\cos\theta}}$

Time period, $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l\cos\theta}{g}}$

if $\theta \to 0$ then $\cos\theta = 1 \rightarrow T = 2\pi\sqrt{\frac{l}{g}}$

## Death Well (a person basically drives a bicycle on vertical surface of large wooden well)




$f = mg$

$N = \frac{mv^2}{r} = m\omega^2 r$

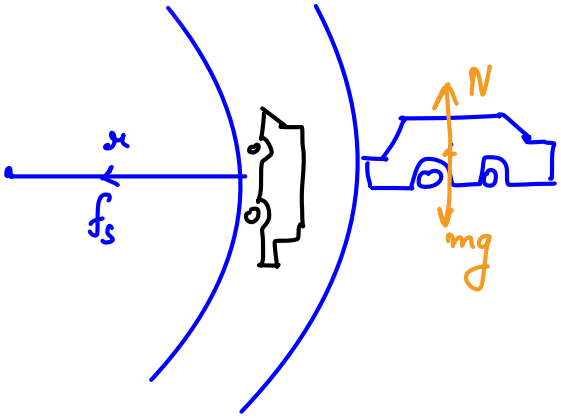
$\mu\left(\frac{mv^2}{r}\right) \geq mg$ , $\mu \geq \frac{rg}{v^2}$

or $v^2 \geq \frac{rg}{\mu}$

## Circular turning of Roads :-

$N = mg$ -(1) and $F_c = f_s$

$f_s = \frac{mv^2}{r}$ -(2)

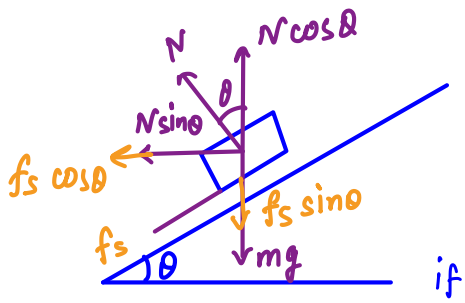



if $f_s = \mu_s N$

then $\frac{mv_{max}^2}{r} = \mu_s N = \mu_s mg$

$\Rightarrow v_{max} = \sqrt{\mu_s rg}$

# Circular motion on Banked Road :




$N\cos\theta = f_s \sin\theta + mg$ — (1)

$\frac{mv^2}{r} = N\sin\theta + f_s\cos\theta$

if $f_s = \mu_s N$ then speed = $v_{max}$

$$\frac{mv_{max}^2}{r} = N\sin\theta + \mu_s N\cos\theta = N[\sin\theta + \mu_s\cos\theta] \quad -(2)$$

from (1) $mg = N\cos\theta - f_s\sin\theta$

$mg = N\cos\theta - \mu_s N\sin\theta$

$mg = N(\cos\theta - \mu_s\sin\theta)$ —(3)

eq (2) ÷ eq (3)

$$\frac{v_{max}^2}{rg} = \left[\frac{\sin\theta + \mu_s\cos\theta}{\cos\theta - \mu_s\sin\theta}\right]$$

Maximum safe speed, $\boxed{v_{max} = \sqrt{rg\left(\frac{\mu_s + \tan\theta}{1 - \mu_s\tan\theta}\right)}}$

If $\mu_s = 0$, $v = \sqrt{rg\tan\theta}$

{ At this speed frictional force is not needed at all to provide the necessary centripetal force }